XIAOYUN YIN

PhD Candidate, Human Systems Engineering

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EDUCATION

Ph.D. in Human Systems Engineering Aug 2022 - May 2027 (expected)

Arizona State University, Polytechnic School, Tempe, AZ

Advisor: Jamie C. Gorman, PhD

Dissertation: Mutual Adaptation in Human-AI Interaction: Modeling Human and AI as a Single Coupled System for Dynamical Research

M.S. in Human Systems Engineering Aug 2020 - Aug 2022

Arizona State University, Tempe, AZ

B.S. in Psychology; B.S. in Sociology (double major) Aug 2015 - May 2020

Arizona State University, Tempe, AZ

Dean's Honors (Fall 2018 - Spring 2019)

RESEARCH INTERESTS

Human-AI mutual adaptation; trust dynamics in human-autonomy teams; cognitive state detection; team cognition; adaptive AI behavior modeling; physiological and behavioral measurement of teamwork; agent-based simulation of human-AI collaboration.

PEER-REVIEWED PUBLICATIONS

Yin, X., Gorman, J. C., Grimm, D. A., Clark, J., Johnson, C. J., Zhou, S., Cooke, N. J., & Amazeen, P. G. (2025). Measuring resilience in human-AI teams under communication delays in space missions. Applied Ergonomics.

Zhou, S., Duan, W., Yin, X., Scalia, M. J., Zhang, R., Weng, N., Schelble, B. G., Funke, G., Tolston, M., Freeman, G., McNeese, N. J., & Gorman, J. C. (2025). [AFOSR Phase 1 study on human-autonomy teaming]. Applied Ergonomics. (Accepted September 2025)

Duan, W., Flathmann, C., McNeese, N., Scalia, M., Zhang, R., Gorman, J., Freeman, G., Zhou, S., Hauptman, A., & Yin, X. (2025). Trusting autonomous teammates in human-AI teams - A literature review. In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems.

Duan, W., Zhou, S., Scalia, M. J., Yin, X., Weng, N., Zhang, R., Freeman, G., Tolston, M., Gorman, J. C., & McNeese, N. J. (2024). [Trust evolvement in human-autonomy teams]. Proceedings of the ACM on Human-Computer Interaction (CSCW).

Huang, L., Freeman, J., Cooke, N. J., Cohen, M. C., Yin, X., Clark, J., ... & Colonna-Romano, J. (2023). Establishing human observer criterion in evaluating artificial social intelligence agents in a search and rescue task. Topics in Cognitive Science.

Cauffman, S. J., Cohen, M. C., Yin, X., Demir, M., Scholcover, F., Huang, L., & Cooke, N. J. (under review). In light of human spatial ability and theory of mind: Considerations for AI-teammate design. Journal of Cognitive Engineering and Decision Making.

CONFERENCE PRESENTATIONS

Yin, X., Zhou, S., Scalia, M. J., Zhang, R., Duan, W., Weng, N., Tuttle, J., Tolston, M., Funke, G., Freeman, G., McNeese, N. J., & Gorman, J. C. (2025). Trust contagion in team of teams (ToT) for human-autonomy teaming. Proceedings of the Human Factors and Ergonomics Society Annual Meeting.

Yin, X., & Gorman, J. C. (2024). Entropy and trust dynamics in human-autonomy teaming. Proceedings of the Human Factors and Ergonomics Society Annual Meeting.

Yin, X., Cooke, N., Gorman, J. C., Grimm, D., Zhou, S., Zahmat Doost, E., Liljenstolpe, M., Pisors, J., & Teo, A. Z. Y. (2024). Resilience and AI integration in dynamic space-based team operations [Lightning talk]. 69th Annual Meeting of HFES.

Yin, X., Zhou, S., Scalia, M. J., Zhang, R., & Gorman, J. C. (2024). Predictive dynamics of trust in human-robot interaction: A recurrent neural network approach. HRI '24 Workshop on Trust in Human-Robot Interaction, Boulder, CO.

Kurukuti, Y., & Yin, X. (2024). Evaluating AMI's effectiveness in measuring emotion dynamics in team interactions [Poster]. 69th Annual Meeting of HFES.

Zhou, S., Yin, X., Scalia, M. J., Zhang, R., Gorman, J. C., & McNeese, N. J. (2023). Development of a real-time trust/distrust metric using interactive hybrid cognitive task analysis. Proceedings of the Human Factors and Ergonomics Society Annual Meeting.

Yin, X., Gorman, J. C., Renshaw, C. E., Cooke, N. J., Grimm, V., & Amazeen, P. G. (2022). Modeling and simulation of teams as complex adaptive systems. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 66(1), 1085-1089.

Yin, X., Clark, J., Johnson, C. J., Grimm, D. A., Zhou, S., Wong, M., Cauffman, S., Demir, M., Cooke, N. J., & Gorman, J. C. (2022). Development of a distributed teaming scenario for future space operations. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 66(1), 788-792.

RESEARCH EXPERIENCE

Research Associate - Systems Psychology Lab Aug 2022 - Present

Center for Human, AI, and Robot Teaming, Arizona State University

Advisor: Jamie C. Gorman, PhD

Lead AFOSR-funded trust contagion project analyzing human-AI team dynamics using multilevel models and dynamic systems analysis.

Developed RNN-based approach for trust prediction; applied entropy-based and information-theoretic metrics (mutual information, transfer entropy, recurrence quantification analysis) to quantify team adaptation in real time.

Developed perturbation-response methodology for measuring team resilience under communication delays.

Active collaborator on the OP TEMPO BioTDMS project - multi-institutional federal grant team spanning ASU, Georgia Tech, Johns Hopkins University, and Charles River Analytics - processing EEG and physiological data for bio-behavioral team dynamics.

Industry collaboration with NirSense (2026): co-design of experimental protocol and IRB documentation for fNIRS-based human-AI teaming study using the NIRSense Aurelian device.

Cross-institutional hyperscanning project with Georgia Tech (Rojas, Zahmat Doost) on n-brain microstate analysis.

Led distributed human-autonomy teaming study across ASU and University of Dayton (AFOSR).

Trained and supervised 10+ undergraduate research assistants.

Graduate Service Assistant - CHART Aug 2020 - Aug 2022

Center for Human, Artificial Intelligence, and Robot Teaming, Arizona State University

Directors: Nancy J. Cooke, PhD & Jamie C. Gorman, PhD

Contributed to DARPA ASIST (Artificial Social Intelligence for Successful Teams) program developing AI models for team cognition; engineered Minecraft testbed for urban search-and-rescue (USAR) human-AI research.

Integrated multimodal data streams (physiological, behavioral, communication) for cognitive state detection.

Collaborated with university library on long-term research data preservation.

Research Assistant - Culture and Decision Science Network Lab Jan 2018 - May 2020

Arizona State University

Statistical analysis of survey data on hacker behaviors using SPSS.

FEDERAL GRANT PARTICIPATION

DARPA ASIST (Artificial Social Intelligence for Successful Teams) - contributing researcher (2020-2023). Developed AI models for team cognition and the Minecraft USAR testbed.

AFOSR Space Challenge - contributing researcher. Investigated distributed space team coordination under variable communication latency.

AFOSR Trust Contagion Project - lead graduate researcher. Multilevel modeling of human-AI team dynamics; ongoing weekly research meetings with PI team.

OP TEMPO BioTDMS Project - multi-institutional federal grant team member; active participant in weekly project meetings with PIs from ASU (Gorman, Cooke), Georgia Tech (M. Li, E. Rojas), Johns Hopkins University, and Charles River Analytics (Bracken, Winder, Desrochers).

NSF STTR - Contact small business company for collaboration, prepare white paper for the proposal.

AWARDS & HONORS

Dieter W. Jahns Student Practitioner Award ($2000 for mentor and mentee) 2026

Principled Engineering Graduate Fellowship, Ira A. Fulton Schools of Engineering, ASU 2026

APA Minority Fellowship Program, Psychology Summer Institute 2026

Conference Travel Grants ($2,000) 2024

Google Cloud Compute Credits ($1,100), Concordia-Contest 2024

HFES Human-AI-Robot Teaming (HART) Technical Group Best Video Award, 1st Place 2021

Engineering Graduate Fellowship, Arizona State University 2020 - 2021

Dean's Honors, Arizona State University 2018 - 2019

TEACHING EXPERIENCE

Guest Lecturer, Arizona State University.

Teaching Assistant & Course Developer, Arizona State University.

Mentored undergraduate students in human-AI teaming research, including the ASU Pallas Fellow program; mentees have transitioned to PhD programs (e.g., Human Factors Psychology PhD at Clemson University, 2026).

Psychology Tutor, Mesa, AZ (2019 - 2020).

APPLICATIONS IN PROGRESS

Summer Institute in AI Methods for Social Scientists (AIMS), 2026 - applicant.

Additional summer school program in computational social science / AI methods, 2026 - applicant.

Predoctoral Editorial Board Reviewer, Translational Issues in Psychological Science (APA), 2026 - applicant.

SERVICE & PROFESSIONAL ACTIVITIES

Peer Review - Journals

International Journal of Human-Computer Interaction (Taylor & Francis) - multiple manuscripts reviewed in 2025-2026, including HIHC-S-2025-2328, HIHC-S-2025-4475 (and revision), HIHC-S-2026-0305, and HIHC-S-2026-1584.

Scientific Reports (Nature Portfolio) - invited reviewer, 2026.

Scientific Data (Nature Portfolio) - invited reviewer, 2026.

Humanities and Social Sciences Communications (Nature Portfolio) - invited reviewer, 2026.

Artificial Intelligence Review (Springer) - invited reviewer, 2026.

Universal Access in the Information Society (Springer) - invited reviewer, 2026.

Applied Ergonomics (Elsevier) - invited reviewer, 2026.

Computational and Mathematical Organization Theory (Springer) - invited reviewer, 2025.

Cognition, Technology & Work (Springer).

Peer Review - Conferences & Workshops

International Conference on Human-Computer Interaction (HCII).

ACM/IEEE International Conference on Human-Robot Interaction (HRI) Workshops.

NeurIPS Main conference & Workshop.

Human Factors and Ergonomics Society (HFES) Annual Meeting.

Conference Service & Organization

Volunteer / Local Organizing committee, HFES Aspire 2024 (hosted at Phoenix, Arizona).

Workshop chair, multi-institutional training workshops for human-AI teaming research, IEEE CogSIMA 2026

Professional Memberships & Affiliations

American Psychological Association (APA), Member.

Human Factors and Ergonomics Society (HFES), Member.

IEEE & IEEE Computational Intelligence Society (CIS), Member.

Women in Big Data, Fellow.

Women in Machine Learning (WiML).

Panels & Mentoring

Panelist, ASU ASpire 2025.

Supervisor, ASU Pallas Fellow undergraduate research program (real-time bio-behavioral team dynamics algorithms).

Graduate-school mentor for undergraduate and incoming PhD students.

INDUSTRY COLLABORATION

NirSense x ASU collaboration (2026): co-investigator on industry-academic study integrating the NIRSense Aurelian fNIRS device into human-AI teaming research. Contributed to experimental design, IRB protocol, and survey instrument development.

TECHNICAL SKILLS

Programming: Python, MATLAB, R, NetLogo, HTML5.

Machine Learning: Deep learning (RNNs), Bayesian modeling, agent-based simulation.

Analysis: Information theory (mutual information, transfer entropy), recurrence quantification analysis, nonlinear dynamics, multilevel modeling, microstate analysis (EEG hyperscanning).

Tools: SPSS, Qualtrics, Minecraft testbed development, fNIRS (NIRSense Aurelian).

Languages: English (fluent), Chinese (native).

RESEARCH IMPACT

Research published in Applied Ergonomics, Proceedings of the ACM on Human-Computer Interaction (CSCW), Proceedings of the 2025 CHI Conference, Proceedings of the Human Factors and Ergonomics Society, and HRI Workshop Proceedings.

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